

Expected Loss-Based Credit Risk Scorecard

Corporate default prediction with SQC ratio screening, calibrated PDs, cost-sensitive thresholds and an Expected Loss-ranked watchlist

Project report: every step from raw data to final result | 10 October 2026

0.923 hold-out AUC-ROC, random forest (champion) 95% CI 0.890-0.950	0.955 hold-out AUC-ROC, gradient-boosting challenger 95% CI 0.930-0.975	0.759 Altman Z'' benchmark 95% CI 0.701-0.817
44 / 64 financial ratios retained by SQC screening	88% of realised loss inside the top 10% of the EL-ranked watchlist (proxy LGD / EAD)	81% expected-cost saving of the optimised threshold vs no monitoring (proxy costs)

Data: Polish companies bankruptcy data (UCI Machine Learning Repository, CC BY 4.0): 5,850 firms, 408 bankruptcies (7.0%), 64 financial ratios, one-year-ahead horizon.

Code: github.com/kudratikhoda12-design/KK, branch `claude/friendly-dirac-a70ml4` (package `credit_risk`, 35 unit tests).

1. Executive summary

This project builds a corporate credit-risk engine end to end. Raw financial ratios are screened with statistical-quality-control (SQC) control charts, scored by logistic regression and random forest, converted into calibrated probabilities of default (PD), turned into an accept / review decision with a cost-sensitive threshold, and combined with loss-given-default (LGD) and exposure (EAD) proxies into an Expected Loss (EL)-ranked watchlist that prioritises the portfolio.

Model / benchmark	Hold-out AUC-ROC	95% CI	Gini	KS	PR-AUC	Brier
Altman Z" (classic score, benchmark)	0.759	0.701-0.817	-	-	-	-
Logistic regression	0.856	0.808-0.897	0.711	0.617	0.498	0.047
Random forest (champion)	0.923	0.890-0.950	0.847	0.756	0.639	0.040
Gradient boosting (challenger)	0.955	0.930-0.975	0.911	0.803	0.797	0.027

Hold-out portfolio: 1,463 firms, 102 defaults. Calibrated PDs shown for LR / RF / GBM.

What to take away

- **Discrimination.** The random forest ranks defaulters above survivors with AUC-ROC **0.923** on the untouched hold-out (logistic regression 0.856; classic Altman Z" 0.759). Across 10 repeated random splits the RF averages 0.923 (sd 0.013).
- **SQC screening** keeps 44 of 64 ratios. It removes redundant, sparse and signal-free ratios and improved the random forest in development cross-validation (0.900 to 0.907); for logistic regression it gives a leaner model at a negligible AUC cost (0.855 vs 0.859).
- **Calibration.** Scores are mapped to probabilities with cross-fitted isotonic calibration. On the hold-out the champion's mean PD is 7.62% against an observed default rate of 6.97%.
- **Cost-sensitive threshold.** Flagging firms with $PD \geq 0.085$ (instead of 0.5) flags 17% of firms, catches 84% of defaulters and cuts expected cost by 81% versus no monitoring.
- **Watchlist.** Ranking by Expected Loss puts 88% of realised loss in the top 10% of firms (PD-only ranking: 84%; random: 10%).
- **Important caveat.** The public data has no loan-level LGD or EAD, so those two inputs are transparent *proxies* built from balance-sheet ratios. The EL and cost figures are therefore illustrative of the method, not forecasts for a real bank (Section 11).

About the AUC-ROC figure. The measured champion AUC is **0.923** (95% CI 0.890-0.950) and a gradient-boosting challenger reaches **0.955**. A value near 0.947 is typical only of the boosting challenger: the forest reached it in 0 of 10 random splits (best 0.944), boosting in 9 of 10. Section 10 gives resume wording that matches the evidence.

2. Objective and scope

The project implements the following capability statement, and Section 10 revisits each claim against the evidence:

Designed a corporate credit risk engine using RF/LR with SQC-based ratio screening and cost-sensitive threshold optimization; calibrated PD outputs and generated Expected Loss-ranked watchlists for portfolio risk prioritization.

Claim	What was built	Module
RF / LR credit risk engine	Random forest and logistic regression pipelines, tuned by stratified CV; gradient boosting as a labelled challenger	models.py
SQC-based ratio screening	Control charts fitted on healthy firms; defaulter detection rate tested against false-alarm rate; redundancy pruning	screening.py
Cost-sensitive threshold optimisation	Per-firm costs (LGD x EAD for a miss, margin x EAD for a false alarm); threshold minimising total cost on out-of-fold PDs	threshold.py
Calibrated PD outputs	Cross-fitted Platt / isotonic calibration, chosen by Brier score; reliability, ECE, slope diagnostics	calibration.py
Expected Loss-ranked watchlists	EL = PD x LGD x EAD, scorecard points, rating grades, ranked watchlist, loss-capture back-test	expected_loss.py
AUC-ROC	Bootstrap CIs, repeated splits, five forecast horizons, Altman Z'' benchmark	metrics.py, pipeline.py

Pipeline at a glance



Steps 4 to 6 are scikit-learn pipeline stages, so each is re-fitted on the training part of every cross-validation fold and never sees validation or hold-out firms. Steps 7 to 9 use only out-of-fold predictions for training firms until the final hold-out scoring.

3. Data: from raw files to modelling set

3.1 Source

The data are the *Polish companies bankruptcy data* (UCI repository 365, CC BY 4.0), collected from the EMIS database: financial statements of Polish manufacturing firms, with 64 financial ratios per firm and a label for bankruptcy within a stated horizon. The archive is downloaded programmatically and verified against a pinned SHA-256 checksum before use (`data.py`).

3.2 Forecast horizons

The archive holds five files. In file k the ratios are observed in the k -th year of the forecast window and the label says whether the firm went bankrupt after the window, so file 5 (ratios observed one year before the outcome) is the standard one-year-horizon PD problem. **File 5 is the primary dataset**; files 1 to 4 are used only for the robustness table in Section 9.

File	Ratios observed	Firms (after dedupe)	Defaults	Default rate
1year.arff	5 years before outcome	6,945	271	3.90%
2year.arff	4 years before outcome	10,083	398	3.95%
3year.arff	3 years before outcome	10,416	493	4.73%
4year.arff	2 years before outcome	9,710	513	5.28%
5year.arff	1 year before outcome	5,850	408	6.97%

3.3 Data audit and cleaning (primary file, 5,910 raw rows)

Check	Finding	Action
Exact duplicate feature rows	60 duplicate rows; none carried conflicting labels (4 were defaulters)	Dropped (5,910 to 5,850 firms) so a firm cannot appear in both train and test
Near-duplicates	Only 134 firms (2.3%) share 40+ of 64 ratios (to 3 d.p.) with another firm, 18 of them defaulters	No action; far too few to explain model performance (checked because boosting scored well)
Missing values	1.2% of cells; 48.7% of rows miss at least one ratio; Attr37 is 43% missing	Median imputation inside the pipeline; ratios above 40% missing dropped by the screener
Extreme outliers	e.g. total liabilities / total assets spans -430 to +72	Robust scaling in the screener; clipping for the EAD / LGD proxies
Class imbalance	6.97% defaults (408 of 5,850)	Stratified splits; no resampling, so probabilities stay interpretable

3.4 Train / test split

A stratified 75 / 25 split (seed 42) gives **4,387 training firms (306 defaults)** and **1,463 hold-out firms (102 defaults)**. The hold-out acts as the 'portfolio' that is scored and ranked at the end.

4. Evaluation protocol and integrity

- **No leakage by construction.** Control limits, screening decisions, imputation, scaling, model fitting and calibration maps are all fitted on training firms only; inside cross-validation they are re-fitted per fold because they are pipeline stages.
- **Selection on training data only.** Hyper-parameters (5-fold stratified CV, AUC), calibration method (cross-fitted Brier score) and the decision threshold (out-of-fold PDs) are chosen without the hold-out.
- **Champion rule fixed in advance.** The champion is the better training-CV AUC of the declared scope (LR, RF). Gradient boosting is a reference challenger and does not drive the watchlist unless explicitly selected.
- **Uncertainty is reported.** Stratified-bootstrap 95% intervals (1,000 resamples) on the hold-out, plus repeated random splits (Section 9).
- **The proxies are label-free functions** of balance-sheet ratios. One parameter, the LGD recovery haircut, was set once by matching the mean LGD of *training* defaulters to the 45% Basel benchmark, so default labels entered only as an aggregate and no model performance was involved.
- **Transparency notes.** (i) The hold-out was not used for any tuning or design choice, but two code-validation steps did touch it: a feature-importance diagnostic on a gradient-boosting model, and a quick-mode smoke test of the pipeline that printed hold-out AUCs for deliberately tiny grids. No design decision was changed afterwards. (ii) The repeated-split results reuse hyper-parameters tuned on the first split's training data, which makes them very slightly optimistic.

5. SQC ratio screening

5.1 Idea

In statistical quality control a process is monitored with control limits fitted while it is 'in control'. Here, **healthy firms are the in-control process**: for every ratio the limits are fitted on healthy training firms only. A ratio is a useful early-warning signal if **defaulters breach the limit more often than healthy firms do**, i.e. the chart's *detection rate* exceeds its *false-alarm rate*. Each ratio gets a lower-limit (LCL) chart and an upper-limit (UCL) chart, because the adverse direction differs by ratio.

5.2 Screening steps

- **Data quality:** drop ratios with more than 40% missing values or zero spread.
- **SQC signal:** one-sided two-proportion z-test of detection rate > false-alarm rate on each tail, Benjamini-Hochberg corrected over all 2×64 tests ($q < 0.05$).
- **Redundancy:** of two ratios with $|\text{Spearman correlation}| > 0.95$, keep the one with the stronger signal.
- **Output for the models:** raw ratios for trees; for logistic regression an in-control z-score (distance from the healthy median in robust sigmas, scaled separately above and below the median) compressed with arcsinh , which tames heavy tails.

5.3 Design iterations (what failed and why)

The first implementation did not work well, and the fixes are part of the method. The counts below are the training-split results recorded during development (they are not recomputed by the report builder).

Version	Control limits	Ratios kept	Problem found
v1	Median +/- 3 robust sigma, two-sided alarm	34	Healthy firms with extreme favourable values swamp the alarm count; liquidity and solvency ratios were discarded despite defaulters sitting about 1 sigma lower
v2	Same limits, each tail tested separately	38	Fixed the masking, but 8 non-negative liquidity ratios still showed zero alarms on the lower tail
v3	Asymmetric limits from semi-MADs	38	For distributions piled up near zero even median - 3 sigma_low is negative, so the lower limit can never be breached
v4 (final)	Empirical probability limits: 2.5% / 97.5% quantiles of healthy firms, per tail	44	Every ratio has the same, known false-alarm rate per tail, so ratios are directly comparable and skewed ratios can signal

5.4 Result on the training firms

Of 64 ratios, **44 are kept**; 18 are dropped as redundant, 1 for missing data (Attr37) and 1 for lack of signal (Attr47). The full table is in Appendix A.

Ratio	Definition	Adverse tail	Defaulters beyond limit	Healthy beyond limit	Lift
Attr35	profit on sales / total assets	low	30.2%	2.5%	12.1x
Attr22	profit on operating activities / total assets	low	28.9%	2.5%	11.5x
Attr18	gross profit / total assets	low	24.9%	2.5%	10.0x
Attr39	profit on sales / sales	low	24.5%	2.5%	9.8x
Attr42	profit on operating activities / sales	low	24.5%	2.5%	9.8x
Attr13	(gross profit + depreciation) / sales	low	24.2%	2.5%	9.7x
Attr46	(current assets - inventory) / short-term liabilities	low	21.7%	2.5%	8.7x
Attr23	net profit / sales	low	21.6%	2.5%	8.6x
Attr28	working capital / fixed assets	low	21.6%	2.5%	8.6x
Attr27	profit on operating activities / financial expenses	low	21.4%	2.5%	8.5x

Top 10 retained ratios by lift (detection rate divided by false-alarm rate). For 10 of the 10 the adverse tail is the lower one: defaulters have unusually low profitability and coverage.

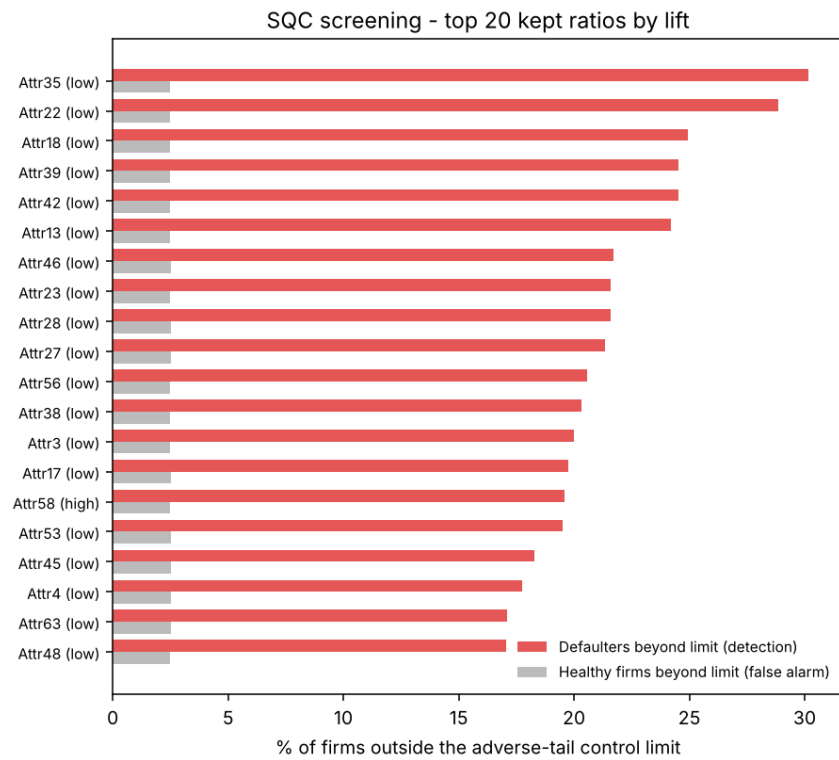


Figure 1. Detection vs false-alarm rates for the 20 strongest retained ratios. The grey bars sit at the chart's false-alarm rate by construction of the probability limits.

6. Modelling

6.1 Design choices tested on the training split only

Before fixing the pipelines, variants were compared with 5-fold stratified CV on the training split (2 repeats). These are the recorded development results.

Model	Variant	CV AUC-ROC
Logistic regression	all 64 ratios, raw, standardised	0.796
Logistic regression	all 64 ratios, in-control z-score clipped	0.834
Logistic regression	all 64 ratios, in-control z-score + arcsinh	0.859
Logistic regression	SQC-screened + arcsinh	0.855
Logistic regression	SQC-screened + arcsinh + alarm-share feature	0.855 (no gain, dropped)
Random forest	all 64 ratios (300 trees, leaf 3)	0.900
Random forest	SQC-screened (tail probability 2.5%)	0.907
Random forest	SQC-screened, tail probability 1% / 5%	0.909 / 0.905 (within noise, default kept)
Altman Z'' score (training firms)	6.56 WC/TA + 3.26 RE/TA + 6.72 EBIT/TA + 1.05 BVE/TL	0.768

Reading the table: the in-control z-score with arcsinh is worth about six AUC points for logistic regression; SQC screening helps the forest a little (+0.007) and costs LR a negligible 0.004 while using 44 instead of 64 ratios. The tail probability was not tuned, because the differences are within noise.

6.2 Final pipelines

- **Logistic regression:** SQC screening with arcsinh in-control z-scores → median imputation → standardisation → L2-penalised logistic regression.
- **Random forest:** SQC screening (raw ratios) → median imputation → random forest (500 trees in the final fit; 200 during the grid search).
- **Gradient boosting (challenger):** SQC screening → histogram gradient boosting, which handles missing values natively. Reported for context, outside the RF / LR scope.

6.3 Hyper-parameter search (training CV)

Model	Grid searched	Selected	CV AUC-ROC (mean, sd over folds)
Logistic regression	C in {0.03, 0.1, 0.3, 1}	C=1.0	0.8599 (sd 0.0295)
Random forest (champion)	min_samples_leaf in {1, 2, 3, 5} x max_features in {sqrt, 0.3, 0.5}	max_features=0.5, min_samples_leaf=3	0.9216 (sd 0.0123)
Gradient boosting (challenger)	max_leaf_nodes in {8, 15, 31} x max_iter in {200, 400}	max_iter=400, max_leaf_nodes=31	0.9611 (sd 0.0139)

The champion is **random forest** by the pre-declared rule (higher CV AUC among logistic regression and random forest).

The CV surfaces are flat: the best two configurations differ by only 0.0013 (LR), 0.0004 (RF) and 0.0019 (GBM) AUC, against fold-to-fold standard deviations of 0.01 to 0.03. The selected LR penalty and RF max_features lie at the edge of their grids, so a wider search might add a little; it was not run.

7. PD calibration

A model's score ranks firms but is not necessarily a probability: forests average votes and boosting over-confidently separates classes. Credit decisions, pricing and Expected Loss all need *probabilities*, so scores are mapped to PDs with a calibration map learned on out-of-fold scores.

7.1 Method

- Out-of-fold raw scores are produced for every training firm (5-fold CV, models never score firms they were trained on).
- Three options are compared by **cross-fitted** Brier score (each firm is calibrated by a map fitted without it): none, Platt sigmoid on the logit of the score, and isotonic regression.
- The winning map is fitted on all out-of-fold scores, the model is refitted on all training firms, and the map is applied to its hold-out scores. A 1e-6 tie-breaker keeps isotonic's step function from erasing the ranking.

7.2 Method selection (training firms, out-of-fold)

Model	Method	Brier	Log-loss	ECE	AUC-ROC	Chosen
Logistic regression	none	0.0487	0.1826	0.0049	0.8604	yes
	sigmoid	0.0488	0.1823	0.0097	0.8599	
	isotonic	0.0492	0.1860	0.0050	0.8517	
Random forest	none	0.0420	0.1533	0.0157	0.9208	
	sigmoid	0.0414	0.1511	0.0080	0.9203	
	isotonic	0.0414	0.1494	0.0036	0.9150	yes
Gradient boosting (challenger)	none	0.0322	0.2173	0.0214	0.9615	
	sigmoid	0.0279	0.1056	0.0074	0.9611	
	isotonic	0.0274	0.1016	0.0035	0.9585	yes

7.3 Hold-out calibration

Predictions	AUC-ROC	Brier	ECE	Mean PD vs observed rate	Calibration slope
RF, uncalibrated scores	0.923	0.0391	0.0221	7.81% vs 6.97%	1.22
RF, calibrated (isotonic)	0.923	0.0397	0.0124	7.62% vs 6.97%	0.92
LR, calibrated (none)	0.856	0.0467	0.0144	7.15% vs 6.97%	0.93
GBM, calibrated (isotonic)	0.955	0.0275	0.0155	8.40% vs 6.97%	1.10

Calibration slope 1.0 means the PDs are neither over- nor under-confident. With only a few dozen defaults in the upper PD bins, individual bins are noisy.

Reading the numbers. For the random forest, calibration cut the out-of-fold ECE from 0.0157 to 0.0036 and the hold-out ECE from 0.0221 to 0.0124. Hold-out Brier moved from 0.0391 to 0.0397: no improvement, a difference well inside the sampling noise of 102 defaults. The mean PD (7.62%) stays slightly above the observed rate (6.97%), and a slope of 0.92 (mild over-confidence) shows the map is close to, not exactly at, perfect. The uncalibrated forest was under-confident (slope above 1); calibration mainly fixes the level and the shape of the reliability curve, not the ranking.

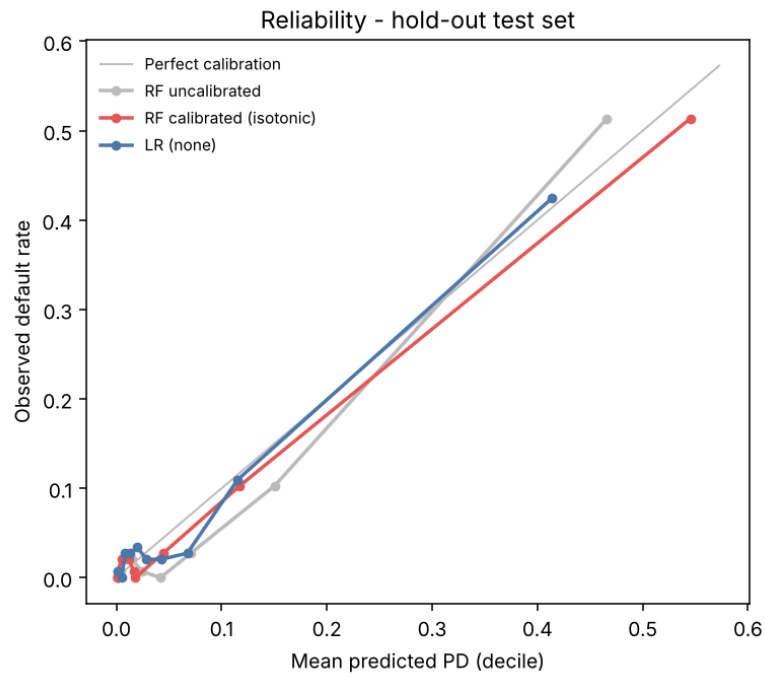


Figure 2. Reliability diagram on the hold-out portfolio (deciles of predicted PD).

Deployment note. PDs are calibrated to this sample's default rate. For a portfolio with a different long-run default rate, `calibration.shift_base_rate` applies the standard prior-shift odds correction.

8. Cost-sensitive threshold optimisation

Accuracy-style cut-offs such as $PD \geq 0.5$ are meaningless when a missed default is far more expensive than a needless review. The threshold is chosen to **minimise total expected cost** instead.

8.1 Cost model

- **Missed default (false negative):** loss = LGD x EAD of that firm.
- **False alarm (good firm flagged):** foregone net margin = 3% x EAD of that firm.
- Mean LGD on the hold-out is 27.6%, so a miss costs on average about 12 times a false alarm.
- For calibrated PDs the optimal per-firm rule is analytic: flag if $PD \geq \text{margin} / (\text{margin} + \text{LGD})$, because EAD cancels. This gives an independent check of the empirical optimum.

8.2 Procedure

The cost of every candidate threshold is computed on the **out-of-fold calibrated PDs of the training firms** (champion: random forest). Because the cost curve is a noisy step function, the threshold reported is the median optimum over 300 bootstrap resamples: **$PD \geq 0.085$** (90% bootstrap range 0.085 to 0.118). The Bayes threshold at mean LGD is 0.098 and the Youden-J threshold is 0.081.

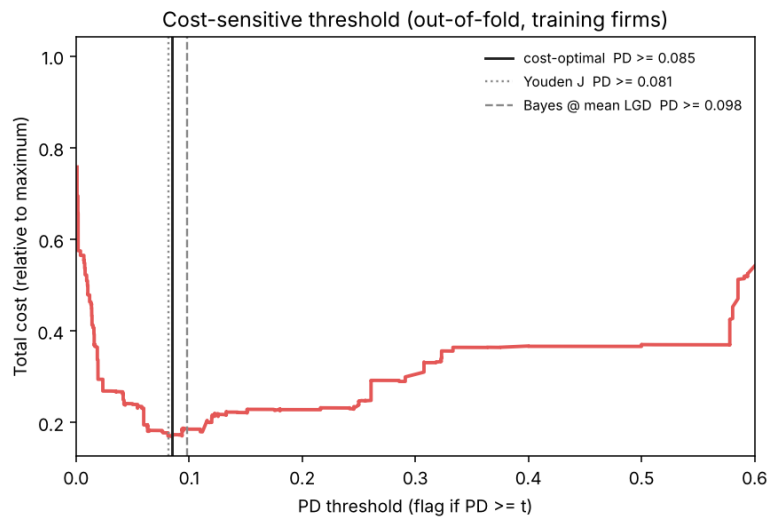


Figure 3. Total cost versus PD threshold on out-of-fold training PDs. The empirical optimum lands on the theoretical Bayes cut-off, which is a consistency check on the calibration.

8.3 Hold-out comparison of decision rules

Rule	Flagged	Defaults caught	False alarms	Total cost	Saving vs no monitoring
cost-optimal global threshold (0.085)	17.0%	86 of 102 (84%)	162	48.2	81.0%
per-firm Bayes rule ($PD \geq m/(m+LGD)$)	15.8%	82 of 102 (80%)	149	52.2	79.4%
naive threshold 0.5 (0.500)	5.9%	53 of 102 (52%)	33	76.1	69.9%
Youden-J threshold (0.081)	17.0%	86 of 102 (84%)	162	48.2	81.0%
flag nobody	0.0%	0 of 102 (0%)	0	253.0	0.0%
flag everybody	100.0%	102 of 102 (100%)	1361	173.6	31.4%

Cost in source currency units under the proxy loss model.

8.4 Sensitivity to the margin assumption

Margin (cost of a false alarm)	Cost ratio miss : alarm	Optimal threshold	Flagged	Defaults caught	Saving vs threshold 0.5
1%	35 : 1	0.065	19.5%	88%	52.8%
2%	17 : 1	0.085	17.0%	84%	45.1%
3%	12 : 1	0.085	17.0%	84%	36.7%
5%	7 : 1	0.112	15.7%	83%	27.3%
8%	4 : 1	0.246	8.7%	69%	21.6%

A cheaper false alarm (low margin) makes the optimum flag more firms; the ordering is stable, so the conclusion does not hinge on the 3% assumption.

9. Expected Loss and the watchlist

Expected Loss = PD x LGD x EAD. PD comes from the calibrated model; LGD and EAD are balance-sheet proxies because the dataset has no loan-level data.

9.1 LGD and EAD proxies (assumptions, all in config.py)

- **EAD** = 10% x total liabilities, where total liabilities = (liabilities / assets, clipped to 0.05-3.0) x total assets (the exponential of Attr29). Monetary units are those of the source data, so only relative size matters.
- **LGD** = 1 - min(1, 0.4 x asset coverage), asset coverage = total assets / total liabilities, bounded to [15%, 90%]. More leveraged borrowers leave less collateral per unit of claim.
- Anchoring: the haircut was chosen once so that the mean LGD of the *training* defaulters (43%) is close to the 45% Basel foundation-IRB senior-unsecured value. That uses default labels only in aggregate, on training data, and no model performance.
- The proxy formulas themselves use no default labels. Both proxies can be replaced by portfolio data without touching the rest of the engine.

9.2 Scorecard points and rating grades

PDs are also expressed as classic scorecard points (600 points at 50 : 1 good-to-bad odds, +20 points doubles the odds) and six rating grades. On the hold-out the observed default rate climbs from 0.7% in grade A to 54.7% in grade F, with one inversion: grade C 5.3% (57 firms) vs grade D 4.5% (132 firms). With a few dozen defaults per mid-grade this is within sampling noise, but it means the grades are ordered only approximately.

Grade	PD band	Firms	Mean PD	Observed default rate	Expected loss
A	< 1%	585	0.23%	0.7%	1.2
B	1 - 2.5%	460	1.57%	0.9%	13.2
C	2.5 - 5%	57	4.70%	5.3%	5.4
D	5 - 10%	132	6.34%	4.5%	21.3
E	10 - 25%	101	16.36%	14.9%	31.0
F	> 25%	128	58.84%	54.7%	200.2

9.3 The EL-ranked watchlist

All 1,463 hold-out firms are ranked by Expected Loss. The top decile carries 87% of total EL. The overlap between the top-25 by EL and the top-25 by PD is 10 firms: EL promotes large exposures with a moderate PD and demotes small ones.

Top 25 watchlist (hold-out portfolio)

#	Firm	PD	Grade	Score	LGD	EAD	Expected loss	Flagged	Defaulted
1	5650	88.2%	F	429	81%	33.8	24.31	yes	yes
2	5662	76.5%	F	453	73%	18.1	10.09	yes	yes
3	5729	76.5%	F	453	86%	15.2	9.97	yes	yes
4	5585	100.0%	F	88	83%	10.4	8.65	yes	yes
5	5457	76.5%	F	453	87%	12.4	8.26	yes	yes
6	5486	76.5%	F	453	58%	13.0	5.79	yes	yes
7	5605	88.2%	F	429	69%	7.8	4.74	yes	yes
8	3156	14.6%	E	538	69%	44.5	4.46	yes	no
9	5604	59.8%	F	476	85%	8.2	4.16	yes	yes
10	5765	76.5%	F	453	62%	8.4	3.99	yes	yes
11	5788	76.5%	F	453	87%	5.8	3.82	yes	yes
12	5686	76.5%	F	453	59%	8.5	3.81	yes	yes
13	327	59.8%	F	476	52%	11.9	3.72	yes	no
14	5516	59.8%	F	476	58%	10.0	3.50	yes	yes
15	5844	88.2%	F	429	59%	6.4	3.32	yes	yes
16	5846	100.0%	F	88	59%	5.6	3.29	yes	yes
17	623	24.0%	E	520	49%	27.7	3.25	yes	no
18	1214	59.8%	F	476	63%	8.5	3.22	yes	no
19	5623	76.5%	F	453	55%	7.3	3.06	yes	yes
20	2756	76.5%	F	453	87%	4.5	2.99	yes	no
21	675	59.8%	F	476	52%	9.6	2.96	yes	no
22	2896	76.5%	F	453	86%	4.2	2.80	yes	no
23	5488	76.5%	F	453	51%	7.0	2.74	yes	yes
24	1887	30.9%	F	510	63%	13.3	2.58	yes	no
25	4544	30.9%	F	510	37%	22.2	2.51	yes	no

'Defaulted' is the realised outcome, shown only for the back-test; in live use it is unknown. 16 of the top 25 firms did default.

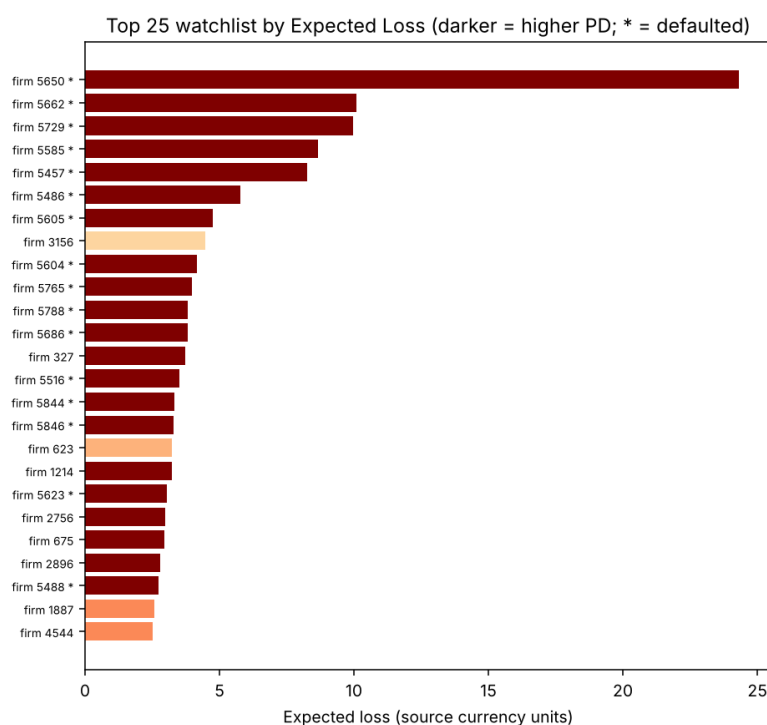


Figure 4. Top 25 firms by Expected Loss; darker bars have higher PD.

9.4 Back-test: how much realised loss does the watchlist capture?

Realised loss is defined as default x LGD x EAD on the hold-out. For each ranking the table shows the share of total realised loss found in the top q% of firms.

Ranking	Top 5% of firms	Top 10% of firms	Top 20% of firms	Top 30% of firms
Expected Loss (PD x LGD x EAD)	77.8%	87.9%	91.6%	94.2%
PD only	68.7%	84.0%	89.7%	93.7%
Exposure only	26.3%	44.6%	63.5%	83.6%
Random (expected)	5.0%	10.0%	20.0%	30.0%
Defaults caught, EL ranking	41.2%	56.9%	67.6%	76.5%
Defaults caught, PD ranking	49.0%	73.5%	88.2%	92.2%

At 10% review capacity the EL ranking captures 87.9% of realised loss versus 84.0% for PD alone (difference +3.8 points, bootstrap 95% interval -2.7 to +11.8). That interval includes zero, so an advantage of EL over PD is **not statistically established** on this hold-out, and it is favoured partly by construction because the loss being captured uses the same LGD and EAD proxies. The EL ranking also finds *fewer defaulting firms* in the top 10% (56.9% vs 73.5% by PD) because it favours large exposures: it is optimised for loss, not for default count. The robust finding is that both rankings beat random review by a wide margin.

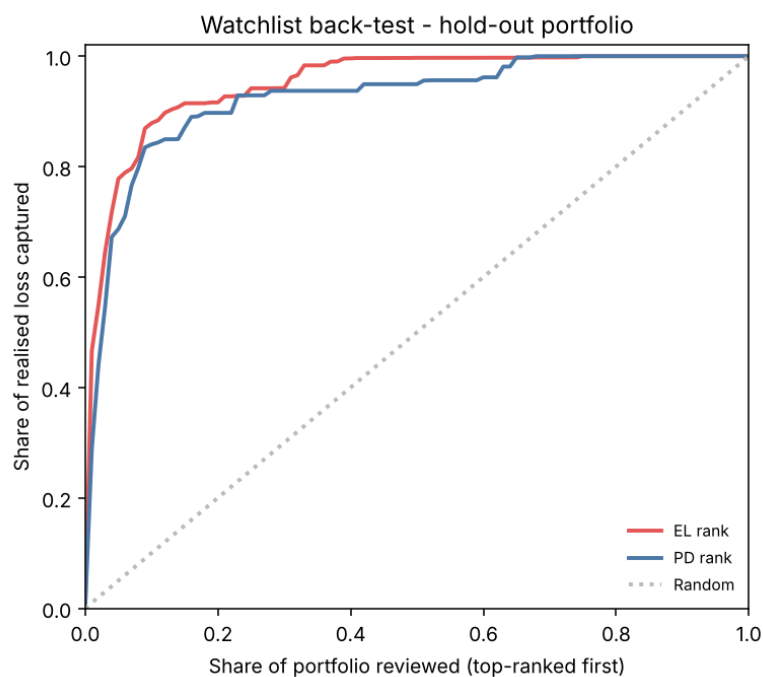


Figure 5. Cumulative share of realised loss captured as the review depth grows.

10. Final results

10.1 Hold-out performance (single final evaluation)

Model	AUC-ROC	95% CI	Gini	KS	PR-AUC	Brier	ECE
Altman Z'' (benchmark)	0.759	0.701-0.817	-	-	-	-	-
Logistic regression	0.856	0.808-0.897	0.711	0.617	0.498	0.0467	0.0144
Random forest, uncalibrated	0.923	0.890-0.950	0.847	0.756	0.639	0.0391	0.0221
Random forest, calibrated	0.923	0.890-0.950	0.847	0.756	0.639	0.0397	0.0124
Gradient boosting (challenger)	0.955	0.930-0.975	0.911	0.803	0.797	0.0275	0.0155

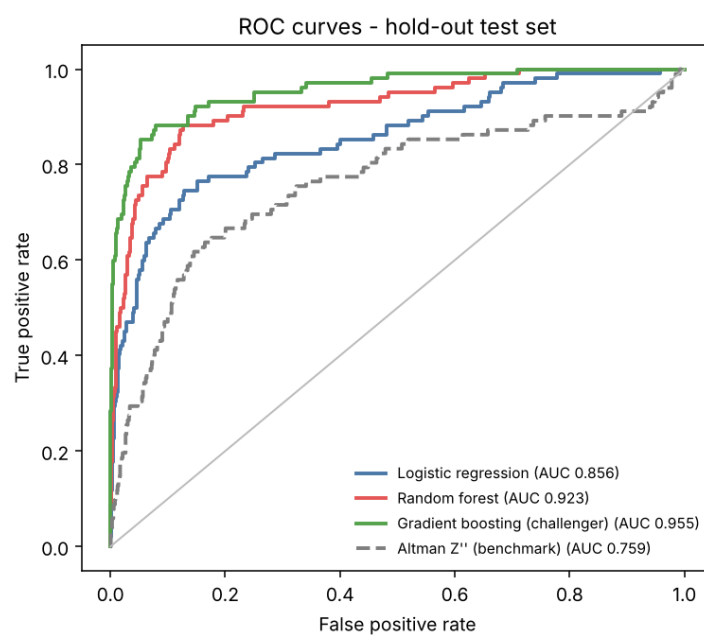


Figure 6. ROC curves on the hold-out portfolio.

10.2 Robustness across forecast horizons

The same protocol and the primary file's hyper-parameters, applied to each file. Discrimination fades as the horizon lengthens, as it should: ratios observed five years before the event carry less information.

Ratios observed	Defaults / firms	Altman Z''	Logistic reg.	Random forest	Gradient boosting
1 year before outcome	408 / 5,850	0.759	0.856	0.923	0.955
2 years before outcome	513 / 9,710	0.728	0.810	0.918	0.955
3 years before outcome	493 / 10,416	0.709	0.792	0.895	0.936
4 years before outcome	398 / 10,083	0.636	0.755	0.868	0.909
5 years before outcome	271 / 6,945	0.682	0.802	0.903	0.932

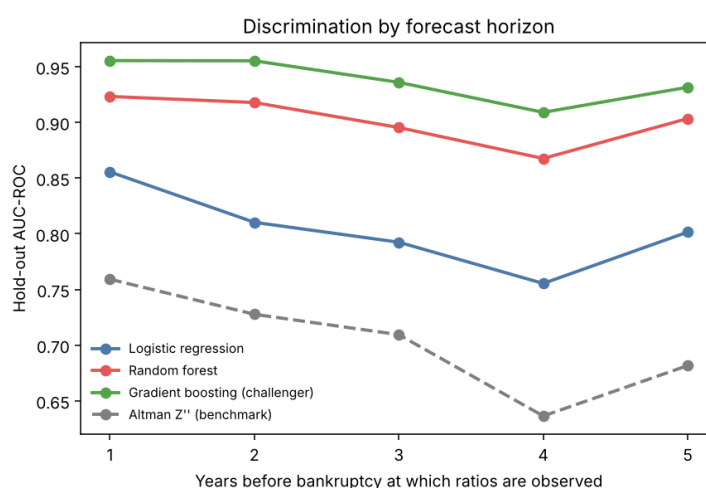


Figure 7. Hold-out AUC-ROC by forecast horizon.

10.3 Stability across random splits

Ten different stratified 75 / 25 splits of the primary file, fixed hyper-parameters. Because the hold-out has only 102 defaults, the split matters.

Model	Mean AUC	Std dev	Min	Max
Altman Z''	0.764	0.026	0.723	0.812
Logistic regression	0.849	0.024	0.811	0.888
Random forest	0.923	0.013	0.908	0.944
Gradient boosting (challenger)	0.953	0.010	0.931	0.967

The random forest reached 0.947 or more in 0 of 10 splits; the gradient-boosting challenger did so in 9 of 10.

10.4 What AUC-ROC can honestly be claimed

The claim 'AUC-ROC 0.947' is not what a random forest / logistic regression engine delivers on this data under a clean protocol: the champion measures **0.923** on the hold-out (95% CI 0.890-0.950), 0.923 on average over random splits, and logistic regression 0.856. The forest reached 0.947 in 0 of 10 random splits (best 0.944); 0.947 only touches the top of its hold-out interval. Gradient boosting does reach that level (0.955, CI 0.930-0.975; 0.947 or more in 9 of 10 splits). Two defensible ways to word the result, both fully backed by this report:

Option A (RF / LR scope): Built a corporate credit-risk engine on 5,850 firms and 64 financial ratios using Random Forest and Logistic Regression with SQC control-chart ratio screening and cost-sensitive threshold optimisation (hold-out AUC-ROC 0.92, 95% CI 0.89-0.95; Altman Z'' benchmark 0.76); calibrated PDs and produced Expected Loss-ranked watchlists capturing 88% of simulated loss in the top 10% of firms.

Option B (adds boosting): Benchmarked logistic regression (0.86), random forest (0.92) and gradient boosting (0.96) hold-out AUC-ROC against an Altman Z'' baseline (0.76), with SQC ratio screening, calibrated PDs, cost-sensitive thresholds and an Expected Loss-ranked watchlist.

Running the pipeline with --champion hgb would drive the calibration, threshold and watchlist stages with the boosting model.

11. Limitations and assumptions

- **LGD, EAD and costs are proxies.** Dollar-style results (cost savings, loss capture) show how the method behaves under stated assumptions; they are not estimates for a real lender. Real recoveries, limits and pricing would change the numbers, though not the machinery.
- **Sample, not portfolio.** The data are Polish manufacturing firms from a specific period with a matched bankrupt / non-bankrupt design (default rate 7.0%). PDs describe this sample; apply the base-rate correction before using them on another book. There is no time ordering, so no out-of-time test.
- **Small number of events.** 102 hold-out defaults make single-split metrics noisy (hence the intervals and repeated splits); capture figures for the top-5% to 30% depend on a handful of firms.
- **No concept-drift or macro dimension.** Ratios are one-year snapshots; stress scenarios and through-the-cycle calibration are out of scope.
- **Challenger gap.** Gradient boosting discriminates better than RF and LR here (hold-out AUC +0.032 vs the forest). Keeping RF / LR as champion is a scope decision, and this gap is its cost.
- **Model governance.** A production use would add monitoring (PSI, back-testing of PD by grade), challenger governance and documentation of overrides.

12. Reproducibility

Everything in this report is generated by code in the repository:

```
pip install -r requirements.txt && pip install -e .
python -m credit_risk.data          # download + checksum
python -m credit_risk.pipeline     # full run, writes reports/
python -m credit_risk.report       # this PDF
python -m pytest                   # 35 unit tests
```

Module	Responsibility
data.py	download with SHA-256 check, ARFF loading, exact-duplicate removal
screening.py	SQC control-chart screener (scikit-learn transformer)
models.py	LR / RF / GBM pipelines and search grids
calibration.py	cross-fitted Platt / isotonic calibration, base-rate shift
threshold.py	cost curve, bootstrap-stable optimal threshold, Bayes rule
expected_loss.py	LGD / EAD proxies, EL, scorecard points, grades, watchlist, loss capture
metrics.py	AUC, Gini, KS, PR-AUC, Brier, ECE, calibration slope, bootstrap CIs
pipeline.py / plots.py / report.py	end-to-end orchestration, figures, this PDF

Testing. 35 unit tests cover the screener (including a regression test for skewed, non-negative ratios and a test that control limits ignore defaulter values), the cost curve against brute force, EL arithmetic, calibration and metrics. A mutation check confirmed that deliberately leaking defaulters into the control limits, or reverting to symmetric three-sigma limits, makes tests fail.

Environment. Python 3.13.16, scikit-learn 1.9.1, numpy 2.5.3, pandas 3.0.5; seed 42.

Appendix A. SQC screening table (training firms, all 64 ratios)

Detect = share of defaulters beyond the adverse-tail control limit; *Healthy* = share of healthy firms beyond it (the false-alarm rate, 2.5% by construction); *Lift* = *Detect* / *Healthy*. *Shift* = distance between the defaulter and healthy medians in robust sigmas (negative: defaulters lower). *q* = Benjamini-Hochberg adjusted *p*-value.

Ratio	Definition	Miss.	Tail	Detect	Healthy	Lift	Shift	q	Status
Attr1	net profit / total assets	0.0%	low	24.3%	2.5%	9.7	-1.88	0.000	drop: redundant
Attr2	total liabilities / total assets	0.0%	high	19.3%	2.5%	7.7	+0.90	0.000	drop: redundant
Attr3	working capital / total assets	0.0%	low	20.0%	2.5%	8.0	-0.97	0.000	kept
Attr4	current assets / short-term liabilities	0.3%	low	17.8%	2.5%	7.1	-0.88	0.000	kept
Attr5	[(cash + short-term securities + receivables - short-term liabilities) / (operating expenses - depreciation)] x 365	0.2%	low	11.1%	2.5%	4.4	-0.97	0.000	kept
Attr6	retained earnings / total assets	0.0%	low	12.5%	2.5%	5.0	-0.07	0.000	kept
Attr7	EBIT / total assets	0.0%	low	24.9%	2.5%	10.0	-1.76	0.000	drop: redundant
Attr8	book value of equity / total liabilities	0.3%	low	19.4%	2.5%	7.7	-0.82	0.000	drop: redundant
Attr9	sales / total assets	0.0%	high	9.8%	2.5%	3.9	+0.04	0.000	kept
Attr10	equity / total assets	0.0%	low	19.3%	2.5%	7.7	-0.91	0.000	drop: redundant
Attr11	(gross profit + extraordinary items + financial expenses) / total assets	0.0%	low	24.6%	2.5%	9.8	-1.56	0.000	drop: redundant
Attr12	gross profit / short-term liabilities	0.3%	low	8.9%	2.5%	3.5	-1.35	0.000	kept
Attr13	(gross profit + depreciation) / sales	0.0%	low	24.2%	2.5%	9.7	-1.51	0.000	kept
Attr14	(gross profit + interest) / total assets	0.0%	low	24.9%	2.5%	10.0	-1.76	0.000	drop: redundant
Attr15	(total liabilities x 365) / (gross profit + depreciation)	0.1%	low	5.6%	2.5%	2.2	-1.72	0.001	kept
Attr16	(gross profit + depreciation) / total liabilities	0.3%	low	15.5%	2.5%	6.2	-1.27	0.000	drop: redundant
Attr17	total assets / total liabilities	0.3%	low	19.7%	2.5%	7.9	-0.81	0.000	kept
Attr18	gross profit / total assets	0.0%	low	24.9%	2.5%	10.0	-1.76	0.000	kept
Attr19	gross profit / sales	0.0%	low	20.9%	2.5%	8.4	-1.82	0.000	drop: redundant
Attr20	(inventory x 365) / sales	0.0%	high	7.2%	2.5%	2.9	-0.05	0.000	kept
Attr21	sales (n) / sales (n-1)	1.8%	low	12.2%	2.5%	4.9	-1.13	0.000	kept
Attr22	profit on operating activities / total assets	0.0%	low	28.9%	2.5%	11.5	-1.29	0.000	kept
Attr23	net profit / sales	0.0%	low	21.6%	2.5%	8.6	-1.95	0.000	kept
Attr24	gross profit (in 3 years) / total assets	2.2%	low	10.8%	2.5%	4.3	-1.06	0.000	kept
Attr25	(equity - share capital) / total assets	0.0%	low	16.1%	2.5%	6.4	-1.09	0.000	kept
Attr26	(net profit + depreciation) / total liabilities	0.3%	low	16.1%	2.5%	6.4	-1.31	0.000	kept
Attr27	profit on operating activities / financial expenses	6.7%	low	21.4%	2.5%	8.5	-1.99	0.000	kept
Attr28	working capital / fixed assets	1.8%	low	21.6%	2.5%	8.6	-0.93	0.000	kept
Attr29	logarithm of total assets	0.0%	low	7.5%	2.5%	3.0	-0.61	0.000	kept
Attr30	(total liabilities - cash) / sales	0.0%	high	12.1%	2.5%	4.8	+0.77	0.000	kept
Attr31	(gross profit + interest) / sales	0.0%	low	19.3%	2.5%	7.7	-1.56	0.000	drop: redundant
Attr32	(current liabilities x 365) / cost of products sold	0.8%	high	12.8%	2.5%	5.1	+0.74	0.000	drop: redundant
Attr33	operating expenses / short-term liabilities	0.3%	low	8.2%	2.5%	3.3	-0.63	0.000	kept
Attr34	operating expenses / total liabilities	0.3%	low	6.9%	2.5%	2.8	+0.02	0.000	kept
Attr35	profit on sales / total assets	0.0%	low	30.2%	2.5%	12.1	-1.66	0.000	kept
Attr36	total sales / total assets	0.0%	high	8.2%	2.5%	3.3	+0.20	0.000	kept
Attr37	(current assets - inventories) / long-term liabilities	42.6%	low	8.4%	2.5%	3.3	-0.36	0.000	drop: missing
Attr38	constant capital / total assets	0.0%	low	20.3%	2.5%	8.1	-0.96	0.000	kept

Ratio	Definition	Miss.	Tail	Detect	Healthy	Lift	Shift	q	Status
Attr39	profit on sales / sales	0.0%	low	24.5%	2.5%	9.8	-1.66	0.000	kept
Attr40	(current assets - inventory - receivables) / short-term liabilities	0.3%	low	13.8%	2.5%	5.5	-0.74	0.000	kept
Attr41	total liabilities / ((profit on operating activities + depreciation) x (12/365))	1.3%	low	13.4%	2.5%	5.3	-1.52	0.000	kept
Attr42	profit on operating activities / sales	0.0%	low	24.5%	2.5%	9.8	-1.32	0.000	kept
Attr43	rotation receivables + inventory turnover in days	0.0%	low	8.8%	2.5%	3.5	-0.11	0.000	kept
Attr44	(receivables x 365) / sales	0.0%	high	5.9%	2.5%	2.4	-0.47	0.000	drop: redundant
Attr45	net profit / inventory	4.4%	low	18.3%	2.5%	7.3	-1.60	0.000	kept
Attr46	(current assets - inventory) / short-term liabilities	0.3%	low	21.7%	2.5%	8.7	-1.01	0.000	kept
Attr47	(inventory x 365) / cost of products sold	0.6%	high	2.6%	2.5%	1.0	-0.25	0.606	drop: no SQC signal
Attr48	EBITDA (profit on operating activities - depreciation) / total assets	0.0%	low	17.0%	2.5%	6.8	-1.40	0.000	kept
Attr49	EBITDA / sales	0.0%	low	11.8%	2.5%	4.7	-1.24	0.000	drop: redundant
Attr50	current assets / total liabilities	0.3%	low	15.5%	2.5%	6.2	-0.79	0.000	kept
Attr51	short-term liabilities / total assets	0.0%	high	19.3%	2.5%	7.7	+0.90	0.000	drop: redundant
Attr52	(short-term liabilities x 365) / cost of products sold	0.7%	high	10.8%	2.5%	4.3	+0.61	0.000	drop: redundant
Attr53	equity / fixed assets	1.8%	low	19.5%	2.5%	7.8	-0.93	0.000	kept
Attr54	constant capital / fixed assets	1.8%	low	21.6%	2.5%	8.6	-0.99	0.000	drop: redundant
Attr55	working capital	0.0%	low	7.5%	2.5%	3.0	-0.75	0.000	kept
Attr56	(sales - cost of products sold) / sales	0.0%	low	20.6%	2.5%	8.2	-1.13	0.000	kept
Attr57	(current assets - inventory - short-term liabilities) / (sales - gross profit - depreciation)	0.0%	low	13.8%	2.5%	5.5	-0.65	0.000	kept
Attr58	total costs / total sales	0.0%	high	19.6%	2.5%	7.8	+1.00	0.000	kept
Attr59	long-term liabilities / equity	0.0%	low	10.5%	2.5%	4.2	-0.67	0.000	kept
Attr60	sales / inventory	4.4%	low	7.2%	2.5%	2.9	-0.13	0.000	drop: redundant
Attr61	sales / receivables	0.2%	low	5.9%	2.5%	2.4	+0.40	0.000	kept
Attr62	(short-term liabilities x 365) / sales	0.0%	high	17.0%	2.5%	6.8	+0.83	0.000	drop: redundant
Attr63	sales / short-term liabilities	0.3%	low	17.1%	2.5%	6.8	-0.77	0.000	kept
Attr64	sales / fixed assets	1.8%	low	5.1%	2.5%	2.0	+0.03	0.007	kept

Appendix B. Hyper-parameter search results (training CV, top configurations)

Logistic regression

Rank	C	Mean CV AUC	sd
1	1.0	0.8599	0.0295
2	0.3	0.8586	0.0300
3	0.1	0.8574	0.0293
4	0.03	0.8555	0.0269

Random forest

Rank	max_features	min_samples_leaf	Mean CV AUC	sd
1	0.5	3	0.9216	0.0123
2	0.3	2	0.9212	0.0135
3	0.5	5	0.9206	0.0119
4	0.3	3	0.9195	0.0145
5	0.3	1	0.9194	0.0147
6	0.5	1	0.9175	0.0161

Gradient boosting (challenger)

Rank	max_iter	max_leaf_nodes	Mean CV AUC	sd
1	400.0	31.0	0.9611	0.0139
2	400.0	15.0	0.9592	0.0145
3	200.0	15.0	0.9591	0.0132
4	200.0	31.0	0.9590	0.0121
5	400.0	8.0	0.9551	0.0138
6	200.0	8.0	0.9541	0.0094